Understanding motion

FORCES ARE ABLE TO TRANSFER ENERGY TO MAKE OBJECTS MOVE.

Forces are rarely applied to an object one at a time and in straight lines. To understand how objects move, some principles are applied.

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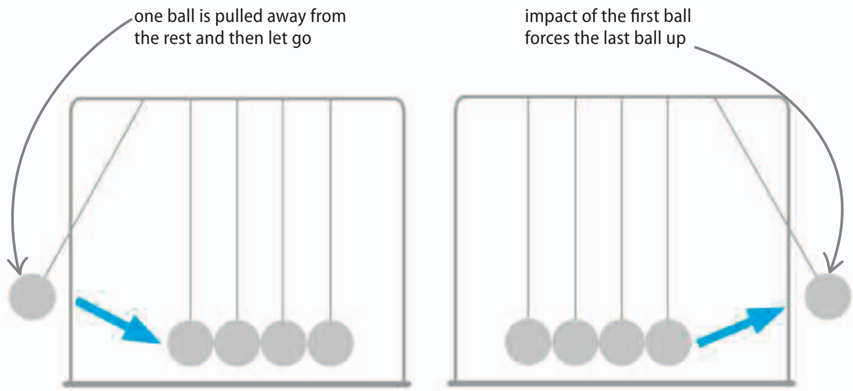
Momentum

A moving object carries on moving because it has momentum. It will keep moving until a force stops it. For example, when you catch a ball, you must exert a force on it to remove its momentum and stop it moving. However, when your hand and the ball collide, the ball will exert a force on your hand so that the momentum of your hand will change. The momentum gained by your hand is equal to the momentum lost by the ball. Momentum is calculated by multiplying the mass of an object by its velocity—the heavier an object and the faster it moves, the greater its momentum. This conservation of momentum, as shown in the illustration, is evidence that energy is never created or destroyed, but transferred between objects.



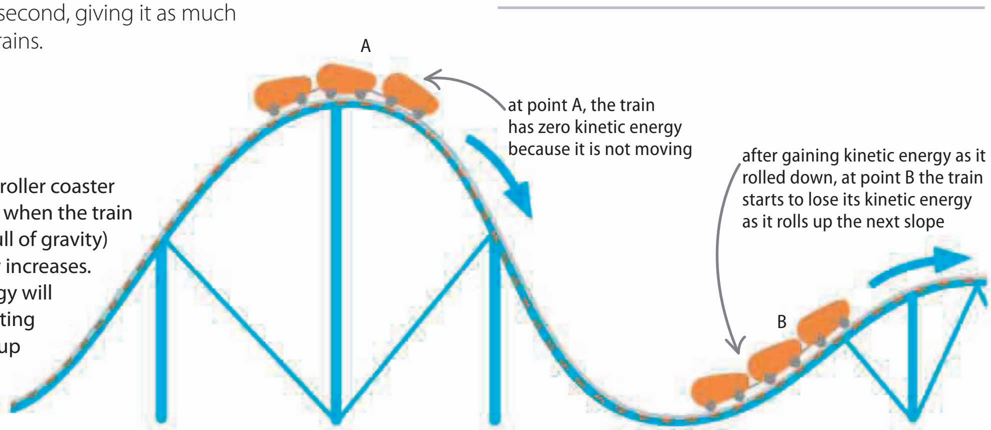
△ **Collision action**
As the left ball hits the line of other balls, its velocity decreases and its momentum drops to zero.

△ **Motion reaction**
The momentum of the first ball passes through its neighbors until it reaches the right ball, which is forced to move.

Kinetic energy

The energy of a moving object is described as kinetic energy. The more kinetic energy an object has, the faster it moves. Some objects can have a relatively small mass but great kinetic energy. For example, the asteroid that is thought to have killed the dinosaurs 65 million years ago had a huge impact despite having a relatively small mass. This is because it hit the ground at around 30 km (19 miles) per second, giving it as much energy as one million express trains.

The **purpose of an engine** is to convert the energy in fuel—or perhaps a battery—into kinetic energy.



▷ **Roller coaster**
At point A, a stationary train on a roller coaster has zero kinetic energy. However, when the train accelerates (in this case, by the pull of gravity) to a high speed, its kinetic energy increases. At point B, the train's kinetic energy will be reduced again by the decelerating effect of gravity as it starts to roll up the smaller slope.